

RailMind AI

Intelligent Railway Safety & Security System

Complete Technical Documentation • v2.0

FAR AWAY 2026 Hackathon • Team Accelerate

Transforming passive CCTV infrastructure into proactive, life-saving intelligence

Domain	Railway Safety & Public Security
Scope	Real-Time Behavioural AI Monitoring
Stack	Python · FastAPI · React · YOLOv8 · LSTM · LangGraph
Team	Team Accelerate — Backend & AI Lead
Event	FAR AWAY 2026 Hackathon
Status	Hackathon Prototype → Production Roadmap

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1. Executive Summary

RailMind AI is an agentic, AI-powered railway safety and security platform that converts existing passive CCTV infrastructure into an intelligent behavioural monitoring network. Using a multi-stage computer-vision pipeline combined with a three-tier agentic reasoning architecture, the system detects high-risk passenger behaviours in real time, calculates dynamic risk scores, and dispatches automated alerts to railway staff — all without installing any new hardware or performing facial recognition.

The platform's core classifier is a bidirectional LSTM network trained on 30-second behavioural sequences, enabling it to capture temporal dependencies that static classifiers miss. This temporal awareness is central to distinguishing genuine distress from ordinary passenger behaviour, significantly reducing false positives.

The platform addresses two critical challenges: suicide prevention and in-station crime (primarily pickpocketing). Designed for rapid deployment via a cloud-based SaaS model, RailMind AI is both scalable and cost-effective, requiring no additional hardware investment.

Metric	Target Value
End-to-end alert latency	< 500 ms
LSTM classification accuracy	> 92% on test set
False positive rate target	< 8%
Hardware requirement	Zero — uses existing CCTV
Supported incident types	Suicide risk, Pickpocketing, Security threats
Deployment model	Cloud SaaS — multi-station

2. Problem Statement

Railway stations are high-density public spaces where critical incidents — suicide attempts, pickpocketing, harassment, and unauthorised track access — can occur with little warning. Despite extensive CCTV coverage, most surveillance systems are passive and rely entirely on human operators to monitor multiple video feeds simultaneously. This makes it extremely difficult to identify subtle behavioural warning signs and respond before an incident occurs.

Core Limitations of Existing Systems

- Human attention fatigue — operators monitoring dozens of feeds simultaneously miss critical cues.
- Reactive posture — current systems record incidents rather than prevent them.
- Subtle signals overlooked — prolonged edge proximity, repetitive pacing, and social withdrawal go undetected.
- Delayed response — by the time an operator notices and dispatches staff, intervention windows are missed.
- Scalability gap — adding more cameras increases the monitoring burden without improving outcomes.

Why This Problem Matters

Impact Area	Consequence
Human Lives	Preventable suicide attempts and assaults occur due to delayed or absent response.
Passenger Safety	Pickpocketing and harassment create unsafe, hostile station environments.
Service Disruption	Incidents cause platform closures, train delays, and cascading network disruptions.
Financial Loss	Emergency responses, legal liability, and disruptions cost railway operators millions annually.
Staff Wellbeing	Operators suffer psychological burden from witnessing preventable tragedies in real time.
Public Trust	Repeated safety failures erode passenger confidence in public rail infrastructure.

The Opportunity

Many incidents could be prevented if suspicious or high-risk behaviours were detected early enough for timely intervention. There is a clear and growing need for an intelligent, scalable, and privacy-preserving solution that can proactively analyse human behaviour, identify potential risks in real time, and assist railway staff before an incident occurs — not after.

3. Project Overview

RailMind AI is an agentic AI-powered railway safety and security system that transforms existing CCTV cameras into intelligent behavioural monitoring agents. The system analyses human movement patterns and detects abnormal or high-risk behaviours in real time using a layered pipeline of computer vision, pose estimation, temporal deep learning, and multi-agent reasoning.

The project addresses two critical railway challenges: suicide prevention and crime detection, particularly pickpocketing and other suspicious activities. Traditional surveillance systems rely on human operators monitoring numerous camera feeds — RailMind AI automates this vigilance, making it continuous, scalable, and consistent.

The solution requires no additional hardware, preserves privacy by avoiding facial recognition entirely, and operates on existing CCTV infrastructure. The LSTM-based temporal classifier is the system's analytical core, enabling it to model behavioural sequences over time rather than evaluating isolated snapshots.

Core Objectives

- Reduce preventable railway fatalities through early behavioural detection and proactive intervention.
- Decrease in-station theft and harassment through continuous, automated monitoring.
- Reduce cognitive load on human operators with AI-assisted triage and prioritisation.
- Provide station managers with actionable analytics, incident history, and hotspot intelligence.
- Enable rapid deployment across railway networks via a SaaS architecture with zero hardware upgrades.
- Maintain strict privacy standards: no facial recognition, no biometric storage, behaviour-only analysis.

4. Key Features

1. Real-Time Behavioural Analysis	Continuously monitors all detected passengers, extracting movement patterns and behavioural features from every video frame with sub-second latency. Processes up to 10 FPS per camera feed.
2. Suicide Risk Detection	Detects distress indicators including prolonged platform-edge proximity (>60s), repetitive pacing cycles, sudden direction reversals, crouching postures, and social withdrawal patterns.
3. Pickpocketing Detection	Identifies suspicious crowd-targeting behaviours: abnormal following distances, repeated close-contact interactions, coordinated group movements near high-density passenger zones.
4. Multi-Agent AI Architecture	Three specialised agents — Perception, Reasoning, and Intervention — work in sequence to detect, evaluate, and act upon identified risks with defined input/output contracts.
5. LSTM Temporal Classifier	Bidirectional LSTM network trained on 30-second behavioural sequences. Captures temporal patterns and trajectory evolution that static classifiers miss, enabling nuanced risk classification.
6. Dynamic Risk Scoring	Risk scores (0-100) computed by combining LSTM classification output, confidence score, behaviour duration, location zone, crowd density, and historical incident patterns.
7. Automated Staff Alerts	Real-time WebSocket notifications dispatched to the nearest available railway staff member, including platform number, risk category, confidence level, and recommended action.
8. Alert Escalation System	Unacknowledged high-risk alerts automatically escalate to supervisors and security personnel after configurable time thresholds (default: 60 seconds).
9. Predictive Hotspot Analytics	Heatmaps and temporal analytics identify platforms, time windows, and spatial zones with disproportionately high incident rates for proactive resource deployment.
10. Privacy-Preserving Design	No facial recognition, no biometric data storage. Track IDs are numeric and session-scoped. All analysis is behaviour-based, GDPR and PDPA compatible.
11. Zero Additional Hardware	Operates entirely on existing CCTV infrastructure. GPU inference handled server-side or on Jetson Orin edge devices. No camera upgrades required.
12. Centralised Monitoring Dashboard	React-based dashboard with live alert feed, platform grid, incident history, heatmaps, analytics charts, and staff response metrics in a single interface.
13. Scalable SaaS Deployment	Cloud-based subscription model enables rapid rollout across multiple stations with centralised management, per-station configuration, and usage-based billing.

5. System Architecture

RailMind AI follows a layered pipeline architecture where each stage processes the output of the previous stage, progressively transforming raw video into actionable intelligence. The architecture is split into an Edge Tier (low-latency CV processing) and a Cloud Tier (reasoning, storage, and dashboard).

Stage	Component	Tier	Description
Input	CCTV Camera Feed	Edge	Existing cameras stream H.264/H.265 via RTSP/ONVIF.
Stage 1	OpenCV Processing	Edge	Decode stream, normalise frames, apply preprocessing filters.
Stage 2	YOLOv8 Detection	Edge	Detect all persons in frame with bounding boxes + confidence.
Stage 3	ByteTrack	Edge	Assign persistent IDs, track individuals across frames.
Stage 4	YOLOv8-Pose	Edge	Extract 17 body keypoints per person per frame.
Stage 5	Feature Extraction	Edge	Compute 7 behavioural features per tracked individual.
Stage 6	LSTM Classifier	Edge	Classify 30-second behavioural sequences into risk categories.
Stage 7	Reasoning Agent	Cloud	LangGraph agent evaluates context, produces final risk score.
Stage 8	Intervention Agent	Cloud	Determines and executes appropriate response actions.
Stage 9	PostgreSQL/SQLite	Cloud	Store incident records, alerts, analytics data.
Stage 10	React Dashboard	Cloud	Live alerts, incident history, heatmaps, analytics.

6. Detailed Workflow

Step 1: Video Capture

Existing CCTV cameras stream video via RTSP or ONVIF protocols. OpenCV captures and decodes each stream into individual frames for processing.

- Protocol: RTSP / ONVIF
- Processing rate: 10 FPS (configurable per camera)
- Resolution: 1920x1080 downscaled to 640x640 for inference
- Output: Normalised BGR frames

Step 2: Person Detection

YOLOv8n/s detects all persons present in each frame with high speed and accuracy, producing bounding boxes for tracking.

- Model: YOLOv8n (speed) or YOLOv8s (accuracy)
- Confidence threshold: 0.45 (configurable)
- Inference time: ~12ms per frame on GPU
- Output: Bounding boxes [x1, y1, x2, y2] + confidence score

Step 3: Multi-Object Tracking

ByteTrack assigns a persistent numeric ID to each detected person and maintains track continuity through occlusions and brief detection gaps.

- Algorithm: ByteTrack — byte-level detection association
- Occlusion handling: Kalman filter prediction during missed detections
- Max track age: 30 frames (3 seconds at 10 FPS)
- Output: Track ID + bounding box per frame

Step 4: Pose Estimation

YOLOv8-Pose extracts 17 COCO body keypoints per tracked individual — enabling analysis of posture, gestures, lean angle, and body orientation.

- Keypoints: Nose, eyes, ears, shoulders, elbows, wrists, hips, knees, ankles
- Output: 17 x [x, y, confidence] per person
- Applications: Head tilt, arm position, body lean, crouching detection, edge-facing orientation

Step 5: Behavioural Feature Extraction

The Feature Extraction module aggregates raw tracking and pose data across a 30-second sliding window to compute a 7-dimensional behavioural feature vector per tracked individual.

- `edge_proximity`: Cumulative seconds within 0.5m of platform edge
- `loitering_time`: Stationary duration in a single spatial zone
- `pacing_count`: Number of back-and-forth movement cycles detected
- `movement_speed`: Average velocity (m/s) over the window
- `direction_changes`: Heading reversals per minute
- `following_distance`: Sustained proximity (metres) to a single other person
- `crowd_interactions`: Count of unique close-contact individuals

Step 6: LSTM Temporal Classification

A bidirectional LSTM network analyses the sequence of feature vectors over the 30-second window to classify behaviour into one of four risk categories. Unlike static classifiers, the LSTM models how behaviour evolves over time.

- Input: Sequence of 30 feature vectors (one per second) — shape [30, 7]
- Architecture: 2-layer Bidirectional LSTM + Dense + Softmax
- Output classes: Normal, Suicide Risk, Pickpocketing Risk, Security Threat
- Output: Class label + confidence score (0.0 - 1.0)
- Inference time: < 5ms on GPU

Step 7: Reasoning Agent

A LangGraph-based Reasoning Agent receives the LSTM classification and evaluates multi-factor context — beyond raw classification — to produce a nuanced final risk score.

- Inputs: LSTM label, confidence score, location zone, behaviour duration, crowd density, historical incident data
- Processing: LangGraph graph workflow with conditional branching
- Output: Risk Score (0-100) + Category + Recommended Action + Reasoning Summary
- Example output: Risk Score 88, Category: Suicide Risk, Action: Approach immediately and offer assistance

Step 8: Intervention Agent

The Intervention Agent executes the appropriate response based on risk score thresholds and monitors acknowledgement to trigger escalation if needed.

- Score 0-39: Silent log only
- Score 40-69: Alert nearest available staff member via dashboard
- Score 70-89: Alert staff + security personnel simultaneously
- Score 90-100: Full emergency escalation + PA system trigger (future v2.0)
- Escalation: Unacknowledged alerts auto-escalate after 60 seconds

7. AI & Computer Vision Pipeline

Model Specifications

Model	Task	Architecture	Input	Latency
YOLOv8n/s	Person Detection	CSPDarknet + PANet + YOLO Head	640x640 image	~12ms
ByteTrack	Multi-Object Tracking	Kalman Filter + Hungarian Algorithm	Detections	~3ms
YOLOv8-Pose	Pose Estimation	YOLOv8 backbone + Keypoint Head	640x640 image	~15ms
BiLSTM Classifier	Behaviour Classification	2-layer Bidirectional LSTM	[30 x 7] seq.	< 5ms
LangGraph Agent	Risk Reasoning	LLM + Conditional Graph Workflow	Structured JSON	~200ms

Feature Vector Definition

Feature	Description	Unit	Window
edge_proximity	Cumulative time within 0.5m of edge	seconds	30s
loitering_time	Stationary duration in single zone	seconds	30s
pacing_count	Back-and-forth movement cycles	count	30s
movement_speed	Average movement velocity	m/s	10s
direction_changes	Heading reversals per minute	count/min	30s
following_distance	Sustained proximity to one individual	metres	30s
crowd_interactions	Unique close-contact individuals	count	30s

8. LSTM Behaviour Classifier — Deep Dive

The LSTM classifier is the analytical heart of RailMind AI. Unlike static classifiers that evaluate a single snapshot of features, the LSTM processes a 30-second temporal sequence, allowing it to model how behaviour evolves over time — which is essential for detecting patterns like gradual edge approach, progressive pacing escalation, and sustained following behaviour.

Why LSTM over Static Classifiers

Criterion	Static Classifier (e.g. XGBoost)	Bidirectional LSTM (RailMind AI)
Input	Single feature vector snapshot	30-second feature sequence [30 x 7]
Temporal awareness	None — no time context	Full — models sequence evolution
Behavioural trajectory	Cannot model progression	Detects gradual escalation patterns
Ambiguous cases	High false positive rate	Context resolves ambiguity
Inference time	< 1ms	< 5ms — still real-time capable
Training complexity	Low	Moderate — requires sequence data
Production suitability	Hackathon prototype	Production-grade

Model Architecture

Input → BiLSTM Layer 1 (128 units) → Dropout (0.3) → BiLSTM Layer 2 (64 units) → Dropout (0.3) → Dense (32, ReLU) → Dense (4, Softmax)

- Input shape: [batch, 30, 7] — 30 timesteps, 7 features
- Bidirectional: processes sequence forward and backward, capturing both historical build-up and future context within the window
- Layer 1 — 128 units: captures low-level temporal feature patterns
- Layer 2 — 64 units: captures higher-level behavioural abstractions
- Dropout 0.3: regularisation to prevent overfitting on limited training data
- Output: softmax probabilities over 4 classes [Normal, Suicide Risk, Pickpocketing, Security Threat]

Training Strategy

- Loss function: Categorical cross-entropy with class weighting (Suicide Risk class weighted 3x due to imbalance)
- Optimiser: Adam (lr=0.001) with ReduceLROnPlateau scheduler
- Data augmentation: Gaussian noise injection on feature sequences to improve generalisation
- Validation: Stratified 80/10/10 train/validation/test split
- Hackathon phase: Trained on synthetic data generated from behavioural rule definitions
- Production phase: Retrain on real incident-labelled data collected from deployed stations
- Target accuracy: >92% on held-out test set

Risk Class Definitions

Class	Trigger Pattern	Threshold
Normal	Typical passenger movement — no sustained and no	Confidence > 0.6
Suicide Risk	Edge proximity >30s + pacing >3 cycles + low crowd interaction	Score > 40
Pickpocketing Risk	Following distance <0.5m sustained + crowd interaction	Score > 45 rapid contacts
Security Threat	High speed + aggressive posture + confrontational	Score > 50

9. Agentic AI Framework

RailMind AI uses a three-tier agentic architecture built with LangGraph. Each agent has a defined role, strict input/output contract, and operates as a node in the LangGraph state graph — enabling conditional branching, parallel processing, and audit-friendly reasoning traces.

Perception Agent

Receives raw detection and tracking data. Runs pose estimation and feature extraction. Assembles the 30-second temporal feature sequence for the LSTM classifier.

- Input: Video frames, YOLOv8 bounding boxes, ByteTrack IDs
- Processing: Pose extraction → per-frame feature computation → 30s sequence assembly → LSTM inference
- Output: {track_id, lstm_label, confidence, feature_sequence, timestamp, location_zone}

Reasoning Agent

Receives the LSTM classification output. Evaluates multi-factor context — behaviour type, confidence, duration, location zone, crowd density, historical patterns — to produce a final risk score with full reasoning trace.

- Input: LSTM label, confidence, feature_sequence summary, location, historical incident data
- Processing: LangGraph conditional graph — context evaluation → score computation → action determination
- Output: {risk_score: 0-100, category, confidence, recommended_action, reasoning_summary}

Intervention Agent

Receives the risk score and recommended action. Executes the appropriate response, creates the incident record, starts the acknowledgement timer, and handles escalation.

- Input: Risk score, category, location, recommended action, staff availability map
- Processing: Threshold evaluation → action selection → WebSocket dispatch → incident creation → escalation timer
- Output: Alert dispatched, incident record written, escalation timer active

10. Tech Stack

Frontend

Library / Tool	Purpose
React.js	Component-based UI for the centralised monitoring dashboard
Tailwind CSS	Utility-first CSS for responsive, accessible design
Recharts	Composable charts for analytics, heatmaps, and trend visualisations
Axios	HTTP client for REST API communication
Socket.io Client	WebSocket client for real-time alert streaming

Backend

Library / Tool	Purpose
Python 3.11+	Core application language
FastAPI	High-performance async REST API framework
Uvicorn	ASGI server for FastAPI
WebSockets	Real-time bidirectional communication for live alerts
Pydantic	Data validation and serialisation for API contracts

Computer Vision

Library / Tool	Purpose
YOLOv8 (Ultralytics)	Person detection — nano/small variant for speed-accuracy balance
YOLOv8-Pose	Human pose estimation — 17 COCO keypoints per person
OpenCV	Video stream capture, decoding, and frame preprocessing
ByteTrack	Multi-object tracking with persistent ID assignment

Deep Learning / ML

Library / Tool	Purpose
PyTorch	LSTM model definition, training, and inference
Bidirectional LSTM	Core behaviour classification model — 30-second sequence analysis
Scikit-learn	Data preprocessing, evaluation metrics, pipeline utilities
NumPy	Numerical computing for feature vector operations

Pandas	Tabular data handling for analytics and training data
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Agentic AI

Library / Tool	Purpose
LangGraph	Graph-based multi-agent orchestration framework
Claude / GPT-4	LLM backbone for the Reasoning Agent (configurable)
LangChain	Prompt management and LLM integration utilities

Database & Infrastructure

Library / Tool	Purpose
SQLite	Hackathon / development — zero-config, local storage
PostgreSQL	Production — robust relational database with full ACID compliance
Redis	Production — pub/sub for WebSocket alerts, caching, rate limiting
Alembic	Database schema versioning and migration management

Communication & Alerts

Library / Tool	Purpose
WebSockets	Primary real-time alert delivery — sub-100ms latency
Email (SMTP)	Secondary alerts for escalated incidents (Level 2+)
SMS (Twilio)	Critical incident alerts for Level 3 escalations

11. Database Design

Core Tables

Table: incidents

```
id, track_id, timestamp, platform_id, camera_id, risk_score, risk_category,  
lstm_confidence, status, resolved_at, resolution_notes
```

Table: alerts

```
id, incident_id, alert_type, sent_at, acknowledged_at, staff_id, escalation_level,  
channel
```

Table: tracks

```
id, track_id, camera_id, session_id, first_seen, last_seen, feature_sequence_json,  
lstm_label, confidence
```

Table: analytics

```
id, date, hour, platform_id, incident_count, avg_risk_score, peak_category,  
false_positive_count
```

Table: staff

```
id, name, platform_zone, contact_email, contact_phone, is_available,  
last_acknowledged_at
```

Table: platforms

```
id, station_id, platform_number, camera_ids_json, edge_zone_config_json,  
incident_count_30d
```

Table: feedback

```
id, alert_id, staff_id, is_false_positive, notes, submitted_at
```

12. API Design (FastAPI)

Method	Endpoint	Description
GET	/api/incidents	List incidents — filter by date, platform, status, category
GET	/api/incidents/{id}	Full incident detail: timeline, features, LSTM output, agent reasoning
POST	/api/incidents/{id}/ack	Staff acknowledges an alert — clears escalation timer
POST	/api/incidents/{id}/feedback	Submit false positive flag or resolution notes
GET	/api/analytics/heatmap	Spatial heatmap data for platform and date range
GET	/api/analytics/summary	Aggregated statistics: daily / weekly / monthly
GET	/api/analytics/lstm-performance	LSTM accuracy, false positive rate, confidence trends
GET	/api/alerts/active	All currently active unacknowledged alerts
POST	/api/alerts/{id}/escalate	Manually escalate an alert to next level
GET	/api/platforms	All monitored platforms and camera status
GET	/api/staff/available	Available staff by platform zone
WS	/ws/alerts	WebSocket stream — real-time alert events
WS	/ws/feed/{camera_id}	WebSocket stream — live annotated video feed

13. Real-Time Communication

RailMind AI uses WebSockets for sub-100ms alert delivery from the Intervention Agent to the dashboard and staff devices. All connected clients receive alerts within one event loop cycle of the Intervention Agent firing.

Alert Payload Structure

```
{ "alert_id": "uuid-v4", "timestamp": "2026-01-15T14:32:07Z", "platform": "Platform 3B", "camera_id": "CAM_003", "risk_score": 88, "risk_category": "Suicide Risk", "lstm_confidence": 0.91, "track_id": 142, "location": {"x": 0.72, "y": 0.41, "zone": "edge"}, "recommended_action": "Approach immediately and offer assistance", "reasoning_summary": "30s edge proximity + pacing x4 + social withdrawal", "escalation_level": 1, "escalate_at": "2026-01-15T14:33:07Z" }
```

Notification Channels

Channel	Trigger Level	Latency	Description
WebSocket Dashboard	All alerts	< 100ms	Primary — real-time alert stream to monitoring dashboard
Mobile Push	Level 1+	< 500ms	In-app notification to nearest available staff device
Email (SMTP)	Level 2+	< 30s	Alert email to supervisor and shift manager
SMS (Twilio)	Level 3	< 60s	Critical SMS to security personnel and station manager

14. Analytics & Dashboard

Dashboard Modules

Live Alert Feed

Real-time scrolling list of active alerts with risk scores, LSTM confidence, platform, and one-click acknowledgement.

Platform Overview Grid

Grid of all monitored platforms showing camera status, active incident count, and current risk level per platform.

Incident History

Searchable, filterable log of all past incidents with full timeline, LSTM output, agent reasoning, and staff response.

Heatmap Visualisation

Spatial heatmaps overlaid on platform diagrams showing incident density by location and time of day.

Temporal Analytics

Line charts of incident trends by hour, day, and week with automatic peak risk window identification.

Risk Category Breakdown

Pie and bar charts showing distribution across Suicide Risk, Pickpocketing, Security Threat categories.

LSTM Performance Panel

Real-time LSTM accuracy, false positive rate, confidence distribution, and per-class precision/recall.

Staff Response Metrics

Average acknowledgement times, resolution rates, per-operator response trends, and false positive submission rates.

15. Privacy & Ethics Framework

Privacy and ethics are foundational design constraints — every architectural decision was evaluated against them. RailMind AI is built to be deployable in any jurisdiction without requiring special biometric data processing exemptions.

No Facial Recognition

Face detection models are explicitly excluded. All analysis uses body movement, posture, and spatial behaviour only. The system cannot identify who a person is.

No Biometric Storage

Track IDs are numeric, session-scoped integers not linked to any identity database. A track ID expires when the session ends and cannot be correlated across sessions.

Minimal Data Retention

Raw video is retained for 72 hours by default. Only flagged incident clips are archived, configurable up to 30 days. Routine footage is not stored.

Behaviour-Only Analysis

The LSTM feature vector contains only anonymised movement metrics. No appearance, demographic, or identity information is processed at any stage.

Human-in-the-Loop

All alerts require human staff confirmation before physical intervention. The AI recommends — humans decide. No automated physical interventions.

Operator Accountability

Every alert, dispatch, acknowledgement, and resolution is immutably logged with a timestamp and operator ID.

Transparent Reasoning

Every incident record includes the full LSTM confidence, risk score computation, and LangGraph reasoning trace for explainability and audit.

Regulatory Alignment

Architecture is designed for GDPR (EU), PDPA (India), and equivalent data protection frameworks. Legal review required per deployment region.

Bias Monitoring

Regular audits of alert rates by platform, time, and incident type to detect and correct potential algorithmic bias in the LSTM classifier.

Public Disclosure

RailMind AI deployment is recommended to be disclosed via station signage, consistent with standard CCTV notification requirements.

16. Edge Deployment Strategy

For real-time railway safety, inference latency is critical. A cloud-only architecture introduces unacceptable network delays and creates single points of failure. RailMind AI uses an edge-cloud hybrid: computationally intensive, latency-sensitive CV stages run on edge devices; reasoning, storage, and dashboard run in the cloud.

Component	Deployment Location	Rationale
YOLOv8 Detection	Edge (Jetson Orin Nano)	Raw video stays local — no bandwidth cost
ByteTrack	Edge	Per-frame continuity requires local state
YOLOv8-Pose	Edge	Latency-sensitive; feeds into feature extraction
Feature Extraction	Edge	Compact output: 7 floats/person vs full video
LSTM Classifier	Edge	TensorRT-optimised; < 5ms inference on Jetson
Reasoning Agent	Cloud	Requires LLM access + historical incident data
Intervention Agent	Cloud	Needs staff DB, notification services, WebSocket hub
Dashboard	Cloud	Multi-station aggregation; browser-accessible
PostgreSQL	Cloud	Centralised storage for cross-station analytics
Redis	Cloud	Pub/sub broker for WebSocket alert distribution

Hardware Specification

Parameter	Specification
Device	NVIDIA Jetson Orin Nano
Cost per unit	~\$500 USD
YOLOv8n FPS	30+ FPS with TensorRT FP16
LSTM inference	< 5ms per sequence
Bandwidth saving	~99.9% vs raw video streaming
Offline buffer	24 hours of local data if cloud connectivity lost
Sync on reconnect	Automatic — feature vectors and incident records only

17. False Positive Handling

Alert fatigue is the leading cause of operator disengagement from AI safety systems. RailMind AI implements a five-layer strategy to keep false positive rates below 8%, preserving operator trust while maintaining high sensitivity for genuine risks.

Layer 1 — LSTM Confidence Thresholding

Alerts are only generated when LSTM confidence exceeds 0.75. Low-confidence predictions are silently logged for retraining data collection but never sent to operators.

Layer 2 — Temporal Confirmation Window

High-risk classification must be sustained for 15+ continuous seconds before triggering an alert. This eliminates single-frame anomalies and transient edge contacts caused by alighting passengers.

Layer 3 — Contextual Suppression

The system learns peak-density time windows (train arrivals, rush hour) from historical data and automatically raises alert thresholds during these periods to suppress expected edge-proximity spikes.

Layer 4 — Operator Feedback Loop

Staff can mark any alert as a False Positive via the dashboard in one click. Labelled false positives are accumulated for weekly LSTM retraining cycles, progressively improving model accuracy.

Layer 5 — Station Performance Monitoring

Automated system alerts flag any station where false positive rate exceeds 40% over a 7-day rolling window. This triggers model recalibration and a review of local edge zone configuration.

18. Improvements & Recommendations

The following improvements are recommended for the production deployment path beyond the hackathon prototype.

Multi-Camera Person Re-Identification

The current system tracks individuals within a single camera's field of view. Production deployment requires re-identifying the same person across multiple camera feeds on a platform.

- Approach: Appearance-based re-ID using lightweight OSNet or FastReID at the cloud tier
- Benefit: Continuity of behavioural tracking across camera handoffs
- Privacy note: Re-ID features are session-scoped and never stored as identity signatures

LSTM Continual Learning Pipeline

The LSTM classifier should improve continuously as labelled incident data accumulates from deployed stations.

- Weekly retraining: Incorporate operator-labelled false positives and confirmed incidents
- A/B shadow testing: New model versions run in parallel with production before promotion
- Versioning: All model versions retained with performance metrics for rollback capability
- Target: 2-3% accuracy improvement per quarter in early deployment phase

Automated PA System Integration

For the highest-risk events (score 90+), an automated PA announcement can be triggered to alert the passenger directly — potentially the fastest intervention possible.

- Implementation: FastAPI endpoint triggers station PA controller via MQTT or REST
- Message: Pre-recorded, calm, non-alarming passenger assistance messages
- Human override: Staff can cancel within 10 seconds before PA fires
- Regulatory requirement: Approval from railway authority required per station

Production Database Migration

SQLite serves the hackathon phase well but has concurrency and scalability limits for multi-station production deployment.

- Phase 1 (Hackathon): SQLite — zero config, sufficient for demo
- Phase 2 (Staging): PostgreSQL + SQLAlchemy ORM — ACID compliant, same query interface
- Phase 3 (Production): PostgreSQL + Redis — Redis handles WebSocket pub/sub and caching
- Migration tool: Alembic for schema versioning and automated migration scripts
- Backup: Daily PostgreSQL dumps + continuous WAL archiving

19. Future Roadmap

v1.0 — Hackathon MVP

- ✓ Recorded video processing pipeline
- ✓ YOLOv8 + ByteTrack + Pose estimation
- ✓ Bidirectional LSTM behaviour classifier
- ✓ Basic LangGraph Perception + Reasoning agents
- ✓ SQLite storage
- ✓ React dashboard with live alerts

v1.5 — Production Alpha

- ✓ Live RTSP stream integration
- ✓ PostgreSQL + Redis migration
- ✓ WebSocket real-time alert delivery
- ✓ Email + SMS notification channels
- ✓ Edge deployment on Jetson Orin Nano
- ✓ Operator false positive feedback system
- ✓ LSTM confidence thresholding + temporal confirmation

v2.0 — Production Release

- ✓ LSTM continual learning pipeline (weekly retraining)
- ✓ Multi-camera person re-identification
- ✓ Automated PA system integration for critical alerts
- ✓ Mobile app for railway staff (React Native)
- ✓ Multi-station SaaS dashboard
- ✓ Compliance reporting and audit log exports
- ✓ Bias monitoring and periodic model audits

v3.0 — Advanced Intelligence

- ✓ Predictive analytics — forecast high-risk time windows before they occur
- ✓ Crowd flow optimisation recommendations for platform managers
- ✓ Integration with emergency services dispatch systems
- ✓ Cross-network anonymised incident pattern sharing
- ✓ Video Swin Transformer for full spatio-temporal modelling
- ✓ Reinforcement learning for dynamic alert threshold optimisation

20. Conclusion

RailMind AI represents a meaningful, technically grounded step forward in public safety technology — transforming the role of CCTV from passive recording to active, intelligent prevention. By combining state-of-the-art computer vision, a temporally-aware LSTM behaviour classifier, multi-agent reasoning, and a privacy-first design philosophy, the system addresses one of the most challenging problems in railway operations: detecting and preventing incidents before they occur.

The LSTM-based temporal classifier is a deliberate architectural choice that distinguishes RailMind AI from simpler rule-based or static ML approaches. Modelling behavioural sequences over time — rather than isolated feature snapshots — is essential for reducing false positives and detecting the gradual escalation patterns that precede most critical incidents.

The architecture is technically sound, the problem domain is genuine and underserved, and the privacy-preserving approach removes a significant barrier to institutional adoption. The improvements outlined — continual learning, multi-camera re-ID, edge deployment, and the ethical framework — provide a clear and credible path from hackathon prototype to production-grade deployment.

With the right investment in data collection, model refinement, and regulatory engagement, RailMind AI has the potential to become a standard component of railway safety infrastructure — making stations safer for passengers, more manageable for operators, and more accountable for railway authorities worldwide.